EXACT DENOMINATOR ASYMPTOTICS, PHASE GEOMETRY, AND MÖBIUS DISJOINTNESS FOR A CLASS OF TWO-STEP TRIANGULAR SKEW PRODUCTS

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Abstract. We study the original Cesàro form of Sarnak's conjecture for the two-step triangular skew product

$$T_h(x,y,z) = (x+\alpha,\ y+\beta\sin(2\pi x),\ z+h(x,y))$$

on $\mathbb{T}^3$, where α is irrational and h is a real-valued finite-band trigonometric polynomial. For every irrational α we obtain an explicit denominator-scale asymptotic formula for the cocycle blocks along the continued-fraction denominators q_j of α, governed by a finite Bessel obstruction vector:

$$S_{q_j}h(x,y) = q_j \sum_n \Gamma_n(h;z_0)\, e^{2\pi i n y} \exp\bigl(i n z_0 \cos\theta_x\bigr) + E_j(x,y).$$

Here $z_0 = \pi\beta/\sin(\pi\{\alpha\})$, and the coefficients $\Gamma_n(h;z_0)$ are given by an explicit finite Bessel expression. If α is Diophantine, then

$$E_j(x,y) = O(q_j^{-1})$$

uniformly, and on the resonant variety $\Gamma_n(h;z_0) = 0$ this yields polynomial-rate rigidity and hence the original Cesàro Möbius disjointness by the criterion of Kanigowski–Lemańczyk–Radziwiłł. The same denominator law also produces a finite-dimensional obstruction quotient with an explicit beta-uniform kernel. It further yields a sectorial rigidity picture for J_0-anchored support-two and support-three profiles. For the pure vertical family

$$z + A\cos(2\pi n_0 y + \varphi),$$

the obstruction reduces to the single scalar equation $J_0(n_0 z_0) = 0$, yielding an explicit infinite resonant family. Finally, the normalized Lommel coefficients form a parity-adapted central-factorial basis in the squared order. This yields exact bounded-bandwidth reconstruction and a Padé reduction of the normalized phase quotient. On the reduced side, the remaining ambiguity is organized by multiplier-incidence strata, while on the realized side the corresponding self-normalized quadratic phase map has an explicitly classified zero-phase fiber. The first bounded windows are governed by a universal top-order quadratic phase relation, whose first shadows are an exact conic ambiguity locus and a degree-1 saturation phenomenon.

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1. Introduction

1.1. **Setup and mechanism.** We study the two-step triangular skew product

$$T_h(x,y,z) = (x+\alpha,\ y+\beta\sin(2\pi x),\ z+h(x,y))$$

on $\mathbb{T}^3$, where α is irrational and h is a real-valued finite-band trigonometric polynomial. The natural denominator blocks are

$$S_q h(x,y) := \sum_{k=0}^{q-1} h(x+k\alpha,\ y+S_k\phi(x)),$$
$$\phi(x) := \beta\sin(2\pi x),$$

so the top cocycle interacts with a nonlinear intermediate layer. The special choice

$$\phi(x) = \beta\sin(2\pi x)$$

turns the denominator-time phase into an exact cosine difference. After a Jacobi–Anger expansion, the whole block becomes an explicit finite Bessel expression. This explicit denominator law is the basic mechanism of the paper.

Write

$$h(x,y) = \sum_{|m|\le M,\ |n|\le N} \widehat{h}(m,n)\, e^{2\pi i m x + n y},$$
$$a := \{\alpha\}, \qquad z_0 := \frac{\pi\beta}{\sin(\pi a)}, \qquad \theta_x := 2\pi x - \pi a.$$

For each vertical frequency n define

$$\Gamma_n(h;z_0) := \sum_{|m|\le M} \widehat{h}(m,n)(-i)^m e^{i\pi m a} J_m(n z_0).$$

1.2. **Results and scope.** The first output is dynamical. For every irrational α, Theorem 4.1 gives the exact denominator asymptotic. If α is Diophantine, Theorem 5.4 upgrades it to a uniform $O(q_j^{-1})$ remainder bound, and on the resonant variety

$$\Gamma_n(h;z_0) = 0$$

Theorem 6.1 yields the original Cesàro Möbius disjointness.

The second output is geometric. Theorem 8.1 identifies a finite-dimensional obstruction quotient and an explicit beta-uniform kernel. On this quotient, Theorem 8.5 gives shared-profile finiteness, Theorem 8.6 closes the J_0-anchored support-two sector, Theorems 8.9 and 8.10 exclude the anchored even-odd and odd-odd support-three sectors at second order, and Theorem 8.12 closes the anchored even-even support-three sector at fifth order. Outside these anchored sectors, Theorem 8.13 shows that the remaining multi-frequency infinite resonance locus is generically thin.

The third output is bounded-bandwidth phase geometry. Theorem 9.1 shows that the normalized Lommel coefficient polynomials form a parity-adapted central-factorial basis in the squared order. This yields the bounded-bandwidth reconstruction theorem Theorem 9.2. On every bounded even/odd bandwidth window, Propositions 9.3 and 9.7 and theorem 9.4 reduce the normalized Prüfer inversion problem to finite-dimensional multiplier-incidence geometry, and Proposition 9.8 identifies the exact universal zero-phase fiber of the realized self-normalized quadratic phase map. The first reduced-side windows can already be computed exactly: Proposition 9.6 gives a circular degree-1 ambiguity locus on $(1,1)$ and full degree-1 saturation from $(1,1)$ into $(2,1)$, both governed by the same universal top-order quadric of Proposition 9.5.

In the pure vertical specialization

$$T_{A,n_0,\varphi}(x,y,z) = (x+\alpha,\ y+\beta\sin(2\pi x),\\ z + A\cos(2\pi n_0 y + \varphi)),$$

Proposition 7.1 and theorem 7.2 reduce the obstruction to the single scalar $J_0(n_0 z_0)$, producing an explicit infinite resonant family.

1.3. **Context and organization.** The closest precedents are the one-step skew-product results of de Faveri, Liu–Sarnak, and Wang [1, 5, 7], together with the rigidity-to-Möbius criterion of Kanigowski, Lemańczyk and Radziwiłł[4]. We use the KLR route, not the complexity route of Huang, Wang and Ye [3]. The genuinely model-specific input is the exact cosine-difference normal form produced by $\phi(x) = \beta\sin(2\pi x)$. This is also why the paper does not claim a comparable theorem for a general intermediate cocycle. The Diophantine assumption enters only at the quantitative stage: the exact denominator law is unconditional, while the Cesàro Möbius theorem uses Diophantine control of the Dirichlet kernels.

Throughout the paper:

1. α is irrational;
2. q_j denotes the continued-fraction denominators of α;
3. in the main Möbius theorems we assume α is Diophantine;
4. h is a real-valued finite-band trigonometric polynomial unless explicitly stated otherwise;
5. $\widehat{h}(0,0) = 0$;
6. $a = \{\alpha\}$, $z_0 = \pi\beta/\sin(\pi a)$, and $\theta_x = 2\pi x - \pi a$.

We fix once and for all a compatible translation-invariant metric on $\mathbb{T}$, still denoted by d, and use the associated product metric on $\mathbb{T}^3$. All sup-norm and rigidity statements are understood with respect to this choice.

The paper follows the proof chain of the denominator law. Section 2 derives the denominator-time normal form for the intermediate cocycle. Section 3 computes the single-mode denominator blocks by a Jacobi–Anger expansion. Section 4 passes from the single-mode calculation to the exact denominator law for finite-band cocycles. Section 5 establishes the quantitative

$O(q_j^{-1})$ remainder bound under a Diophantine condition, derives resonant polynomial-rate rigidity, and verifies the precise Kanigowski–Lemańczyk–Radziwiłłpolynomial-rigidity hypothesis. Section 6 uses that rigidity input to prove Theorem 6.1. Section 7 treats the pure vertical specialization. Section 8 returns to the obstruction quotient and develops the support-two and support-three rigidity results together with the generic finite-resonance theorem. Section 9 develops the central-factorial basis, bounded-bandwidth reconstruction, the reduced-side multiplier-incidence geometry, and the realized quadratic phase geometry on bounded windows.

2. Denominator-time normal form

We begin by rewriting the intermediate cocycle at denominator times. The point of this section is that the special sinusoidal form of ϕ turns the denominator-time phase into a cosine difference, which is the normal form used throughout the remainder of the paper.

Proposition 2.1. *For every $k \geq 0$ and $x \in \mathbb{T}$,*

$$\begin{aligned} S_k\phi(x) &= \beta\,\Im\left(e^{2\pi i x}\frac{1-e^{2\pi i k\alpha}}{1-e^{2\pi i\alpha}}\right) \\ &= \frac{z_0}{2\pi}\big(\cos\theta_x - \cos(\theta_x + 2\pi\{k\alpha\})\big). \end{aligned}$$

Consequently, for every integer n,

$$\begin{aligned} e^{2\pi i n S_k\phi(x)} &= \exp\big(i n z_0 \cos\theta_x\big) \\ &\qquad \times \exp\big(-i n z_0 \cos(\theta_x + 2\pi\{k\alpha\})\big). \end{aligned}$$

Proof. The geometric-series identity gives

$$\sum_{r=0}^{k-1} e^{2\pi i x + r\alpha} = e^{2\pi i x}\frac{1-e^{2\pi i k\alpha}}{1-e^{2\pi i\alpha}}.$$

Taking imaginary parts and using $\sin(2\pi u) = \Im(e^{2\pi i u})$ proves the first identity. Next,

$$1 - e^{2\pi i t} = -2i\, e^{\pi i t}\sin(\pi t),$$

so

$$\frac{1-e^{2\pi i t}}{1-e^{2\pi i\alpha}} = e^{\pi i(t-a)}\frac{\sin(\pi t)}{\sin(\pi a)}.$$

Substituting $t = \{k\alpha\}$ yields

$$S_k\phi(x) = \beta\frac{\sin(\pi\{k\alpha\})\sin(\theta_x + \pi\{k\alpha\})}{\sin(\pi a)}.$$

Using $2\sin u\sin v = \cos(v-u) - \cos(u+v)$ gives the cosine-difference form. Exponentiating gives the last identity. □

3. Single-mode denominator blocks

Once the intermediate cocycle is in normal form, the top cocycle can be analyzed one Fourier mode at a time. The computation in this section isolates the diagonal term responsible for the main denominator contribution and separates it from the off-diagonal remainder.

Proposition 3.1. *For the Fourier mode*

$$h_{m,n}(x,y) = e^{2\pi i m x + n y},$$

one has

$$\begin{aligned} S_q h_{m,n}(x,y) = e^{2\pi i m x + n y} &\exp\bigl(i n z_0 \cos\theta_x\bigr) \\ &\times \sum_{r\in\mathbb{Z}} (-i)^r J_r(n z_0) e^{i r \theta_x} D_q((m+r)\alpha), \end{aligned}$$

where

$$D_q(\xi) := \sum_{k=0}^{q-1} e^{2\pi i k \xi}.$$

In particular, at $q = q_j$,

$$\begin{aligned} S_{q_j} h_{m,n}(x,y) = q_j (-i)^m e^{i\pi m a} J_m(n z_0)\, e^{2\pi i n y} &\exp\bigl(i n z_0 \cos\theta_x\bigr) \\ &+ R_j^{m,n}(x,y), \end{aligned}$$

where

$$\begin{aligned} R_j^{m,n}(x,y) = e^{2\pi i m x + n y} &\exp\bigl(i n z_0 \cos\theta_x\bigr) \\ &\times \sum_{r\neq -m} (-i)^r J_r(n z_0) e^{i r \theta_x} D_{q_j}((m+r)\alpha). \end{aligned}$$

Proof. By definition,

$$S_q h_{m,n}(x,y) = \sum_{k=0}^{q-1} e^{2\pi i m (x+k\alpha) + n(y + S_k\phi(x))}.$$

Use Proposition 2.1 to write

$$e^{2\pi i n S_k \phi(x)} = \exp\bigl(i n z_0 \cos\theta_x\bigr) \exp\bigl(-i n z_0 \cos(\theta_x + 2\pi k \alpha)\bigr).$$

Apply the Jacobi–Anger expansion

$$\exp(-i u \cos v) = \sum_{r\in\mathbb{Z}} (-i)^r J_r(u) e^{i r v}$$

with $u = n z_0$ and $v = \theta_x + 2\pi k\alpha$. Interchanging the finite sum in k with the absolutely convergent Bessel series gives

$$\begin{aligned} S_q h_{m,n}(x,y) = e^{2\pi i m x + n y} &\exp\bigl(i n z_0 \cos\theta_x\bigr) \\ &\times \sum_{r\in\mathbb{Z}} (-i)^r J_r(n z_0) e^{i r \theta_x} D_q((m+r)\alpha), \end{aligned}$$

which is the claimed factorization formula. At $q = q_j$ the unique diagonal index is $r = -m$. Since $D_{q_j}(0) = q_j$ and $J_{-m}(u) = (-1)^m J_m(u)$, the diagonal contribution is

$$q_j(-i)^{-m}(-1)^m J_m(nz_0)\, e^{2\pi imx+ny} e^{-im\theta_x}\, \exp\!\big(inz_0 \cos\theta_x\big).$$

Finally,

$$e^{2\pi imx} e^{-im\theta_x} = e^{2\pi imx} e^{-im(2\pi x-\pi a)} = e^{i\pi ma},$$

so the diagonal term becomes the main term stated in the proposition. □

4. Exact denominator law

We now pass from the single-mode factorization to the full finite-band cocycle. The key point is that, at denominator times, each Fourier mode contributes one diagonal term and an explicit off-diagonal remainder. Summing the diagonal terms produces the obstruction vector.

Theorem 4.1. *Let*

$$\Gamma_n(h; z_0) := \sum_{|m|\le M} \widehat{h}(m,n)(-i)^m e^{i\pi ma} J_m(nz_0).$$

Then

$$S_{q_j} h(x,y) = q_j \sum_{|n|\le N} \Gamma_n(h; z_0)\, e^{2\pi iny} \exp\!\big(inz_0 \cos\theta_x\big) + E_j(x,y),$$

where

$$E_j(x,y) := \sum_{|m|\le M,\ |n|\le N} \widehat{h}(m,n) R_j^{m,n}(x,y).$$

Proof. Expand h into its finitely many Fourier modes and apply Proposition 3.1 to each mode. For each pair (m,n), the diagonal contribution arises from the unique index $r = -m$ and therefore contributes

$$q_j \widehat{h}(m,n)(-i)^m e^{i\pi ma} J_m(nz_0)\, e^{2\pi iny}\, \exp\!\big(inz_0 \cos\theta_x\big).$$

Summing these diagonal terms over the finitely many Fourier modes of h yields the principal part

$$q_j \sum_{|n|\le N} \Gamma_n(h; z_0)\, e^{2\pi iny} \exp\!\big(inz_0 \cos\theta_x\big).$$

All terms with $r \neq -m$ are, by definition, absorbed into E_j. □

Remark 4.2. The statement of Theorem 4.1 requires only that α be irrational. No Diophantine input is used up to this point.

Remark 4.3. Two basic consequences of Theorem 4.1 will be used repeatedly.

1. The zero vertical frequency is the mean:
$$\Gamma_0(h; z_0) = \widehat{h}(0,0).$$
Hence the standing assumption $\widehat{h}(0,0) = 0$ is the vanishing of the $n = 0$ obstruction.
2. If h is real-valued, then
$$\Gamma_{-n}(h; z_0) = \overline{\Gamma_n(h; z_0)}$$
for every $n \in \mathbb{Z}$. Thus the obstruction vector is determined by its positive vertical frequencies.

5. Quantitative remainder bounds and resonant rigidity

For the rest of Sections 5–7, assume that α is Diophantine:
$$\|s\alpha\| \geq c_\alpha \, |s|^{-\tau}$$
for all nonzero integers s.

The goal of this section is to convert the exact obstruction formula of Theorem 4.1 into a quantitative denominator-scale rigidity statement. The argument has two ingredients. The first is a Diophantine bound for the Dirichlet kernels $D_{q_j}(s\alpha)$ on the range $|s| < q_{j+1}$. The second is the rapid decay of Bessel coefficients in the order, which controls the complementary tail. Their combination yields an $O(q_j^{-1})$ remainder estimate for the full finite-band cocycle and therefore polynomial-rate rigidity on the resonant variety.

Lemma 5.1. *For every nonzero integer s and every j:*

1. *if $0 < |s| < q_{j+1}$, then*
$$\left|D_{q_j}(s\alpha)\right| \leq C_\alpha \, |s|^{\tau+1} \, q_{j+1}^{-1};$$
2. *for arbitrary s,*
$$\left|D_{q_j}(s\alpha)\right| \leq q_j.$$

Proof. Use
$$D_{q_j}(s\alpha) = \frac{\sin(\pi q_j s\alpha)}{\sin(\pi s\alpha)}.$$
If $0 < |s| < q_{j+1}$, then
$$\|q_j s\alpha\| \leq |s| \, \|q_j \alpha\| \leq |s| \, q_{j+1}^{-1},$$
so
$$|\sin(\pi q_j s\alpha)| \leq \pi \, |s| \, q_{j+1}^{-1}.$$
The Diophantine condition gives
$$|\sin(\pi s\alpha)| \geq c'_\alpha \, \|s\alpha\| \geq c''_\alpha \, |s|^{-\tau}.$$
Combining the estimates proves the first bound. The second follows immediately from the definition of D_{q_j} as a sum of q_j unimodular terms. □

Lemma 5.2. *Fix $u \in \mathbb{R}$. Then for every integer r,*

$$|J_r(u)| \le C_u \frac{(|u|/2)^{|r|}}{|r|!}.$$

Consequently, for every $A > 0$ there exists $C_{u,A}$ such that

$$\sum_{|r|\ge L} |J_r(u)| \le C_{u,A} L^{-A}$$

for all integers $L \ge 1$.

Proof. For $r \ge 0$, the series definition of J_r gives

$$J_r(u) = \sum_{\ell\ge 0} \frac{(-1)^\ell (u/2)^{2\ell+r}}{\ell!(\ell+r)!},$$

hence

$$|J_r(u)| \le \left(\frac{|u|}{2}\right)^r \sum_{\ell\ge 0} \frac{(|u|/2)^{2\ell}}{(\ell!)^2} \le C_u \frac{(|u|/2)^r}{r!}.$$

For negative indices use $J_{-r}(u) = (-1)^r J_r(u)$. The tail bound follows because factorial decay dominates every power. □

Proposition 5.3. *For each fixed pair (m, n),*

$$\sup_{x,y} \left| R_j^{m,n}(x,y) \right| \le C_{m,n,\alpha,\beta}\, q_j^{-1}.$$

Proof. Since the outer phase factors have unit modulus,

$$\sup_{x,y} \left| R_j^{m,n}(x,y) \right| \le \sum_{r\ne -m} |J_r(nz_0)| \left| D_{q_j}((m+r)\alpha) \right|.$$

Set $s = m + r$ and split into $|s| < q_{j+1}$ and $|s| \ge q_{j+1}$. For the first range, Lemma 5.1 gives

$$\sum_{0<|s|<q_{j+1}} |J_{s-m}(nz_0)| \left| D_{q_j}(s\alpha) \right| \le C_\alpha q_{j+1}^{-1} \sum_{s\ne 0} |s|^{\tau+1} |J_{s-m}(nz_0)|,$$

and the series on the right converges by Lemma 5.2. Thus the contribution of this range is at most $C'_{m,n,\alpha,\beta} q_{j+1}^{-1}$. For the tail use

$$q_j \sum_{|s|\ge q_{j+1}} |J_{s-m}(nz_0)|,$$

and apply Lemma 5.2 with $A = 2$ to obtain

$$q_j \sum_{|s|\ge q_{j+1}} |J_{s-m}(nz_0)| \le C''_{m,n,\beta} q_j q_{j+1}^{-2} \le C''_{m,n,\beta} q_j^{-1}.$$

Combining the two ranges proves the claim. □

Theorem 5.4. *There exists $C(h, \alpha, \beta)$ such that*

$$\sup_{x,y} |E_j(x,y)| \le C(h,\alpha,\beta)\, q_j^{-1}$$

for all j.

Proof. The sum defining E_j contains only finitely many pairs (m,n), so Theorem 4.1 and Proposition 5.3 imply

$$\begin{aligned}\sup_{x,y}|E_j(x,y)| &\le \sum_{|m|\le M,\ |n|\le N}\left|\widehat{h}(m,n)\right| \sup_{x,y}\left|R_j^{m,n}(x,y)\right| \\ &\le C(h,\alpha,\beta)\, q_j^{-1},\end{aligned}$$

where the final constant is obtained by summing the finitely many modewise constants. □

Corollary 5.5. *If*

$$\Gamma_n(h;z_0)=0$$

for every active vertical frequency n, then

$$\sup_{x,y}\left|S_{q_j}h(x,y)\right| \le C(h,\alpha,\beta)\, q_j^{-1}.$$

Proof. Under the resonant hypothesis, the principal part in Theorem 4.1 vanishes identically, so Theorem 5.4 gives the estimate. □

Corollary 5.6. *Under the assumptions of Corollary 5.5,*

$$\sup_{(x,y,z)\in\mathbb{T}^3} d(T_h^{q_j}(x,y,z),(x,y,z)) \le Cq_j^{-1}.$$

Proof. The x-coordinate displacement is $\|q_j\alpha\|\le q_j^{-1}$. For the y-coordinate, Proposition 2.1 gives

$$\sup_x\left|S_{q_j}\phi(x)\right| \le |\beta|\ \frac{\left|1-e^{2\pi i q_j\alpha}\right|}{\left|1-e^{2\pi i\alpha}\right|} \le C(\alpha,\beta)\ \|q_j\alpha\| \le C(\alpha,\beta)\, q_j^{-1},$$

while the z-coordinate displacement is controlled by Corollary 5.5. Since the metric on $\mathbb{T}^3$ is the product metric, the three coordinate estimates combine to give the claimed bound. □

Corollary 5.7. *Let F be a trigonometric polynomial on $\mathbb{T}^3$. Then there exists C_F such that*

$$\left\|F\circ T_h^{q_j}-F\right\|_\infty \le C_F q_j^{-1}.$$

Moreover, for every δ with $0<\delta<2/3$,

$$\sum_{|r|\le q_j^\delta}\left\|F\circ T_h^{rq_j}-F\right\|_\infty^2 \to 0.$$

The same conclusion holds in $L^2(\nu)$ for every T_h-invariant Borel probability measure ν.

Proof. Trigonometric polynomials are Lipschitz. By Corollary 5.6,

$$\left\|F\circ T_h^{q_j}-F\right\|_\infty \le C_F q_j^{-1}.$$

For $|r|\le q_j^\delta$, telescoping gives

$$\left\|F\circ T_h^{rq_j}-F\right\|_\infty \le C_F\,|r|\,q_j^{-1} \le C_F q_j^{\delta-1}.$$

Hence

$$\sum_{|r|\le q_j^\delta} \left\| F\circ T_h^{rq_j} - F \right\|_\infty^2 \le (2q_j^\delta+1)C_F^2 q_j^{2\delta-2}$$
$$\le C_F' q_j^{3\delta-2},$$

which tends to 0 when $\delta < 2/3$. The $L^2(\nu)$ statement follows from the sup norm bound. □

6. Möbius disjointness on the resonant variety

At this stage the dynamical input is complete: on the resonant variety, the cocycle blocks are quantitatively rigid along the denominator sequence. This is the hypothesis needed to invoke the Kanigowski–Lemańczyk–Radziwiłłcriterion.

Theorem 6.1. *Assume that α is Diophantine and that*

$$\Gamma_n(h; z_0) = 0$$

for every active vertical frequency n. Then the system T_h satisfies the original Cesàro Sarnak conjecture.

Proof. The resonant hypothesis implies Corollary 5.5, hence also Corollaries 5.6 and 5.7. Fix a T_h-invariant Borel probability measure ν, and let $\mathcal{F}_{\text{trig}}$ denote the trigonometric polynomials on $\mathbb{T}^3$. Then for every $F \in \mathcal{F}_{\text{trig}}$ and every $\delta \in (0, 2/3)$ the denominator sequence q_j satisfies

$$\sum_{|r|\le q_j^\delta} \left\| F\circ T_h^{rq_j} - F \right\|_{L^2(\nu)}^2 \to 0.$$

Since $\mathcal{F}_{\text{trig}}$ is linearly dense in $C(\mathbb{T}^3)$, this is exactly the PR-rigidity condition of [4, Definition 1.1] for the measure-theoretic system $(\mathbb{T}^3, \mathcal{B}, \nu, T_h)$. As ν was arbitrary, the topological system $(\mathbb{T}^3, T_h)$ is good in the sense of [4, Definition 1.1]. Therefore [4, Theorem 1.1] applies and yields the Möbius disjointness of T_h, i.e. the original Cesàro formulation. □

7. Pure vertical specialization

We conclude by specializing the obstruction formula to a one-frequency family for which the finite obstruction vector collapses to a single Bessel factor. Because J_0 has infinitely many positive zeros, this produces an infinite explicit resonant family.

For the pure vertical family

$$h(x, y) = A\cos(2\pi n_0 y + \varphi),$$

the obstruction vector can be computed explicitly.

Proposition 7.1. *Let*

$$h(x,y) = A\cos(2\pi n_0 y + \varphi).$$

Then

$$\Gamma_{n_0}(h;z_0) = \frac{A}{2}e^{i\varphi}J_0(n_0 z_0), \qquad \Gamma_{-n_0}(h;z_0) = \frac{A}{2}e^{-i\varphi}J_0(n_0 z_0),$$

and

$$\Gamma_n(h;z_0) = 0$$

for every $n \notin \{\pm n_0\}$.

Proof. Write

$$h(x,y) = \frac{Ae^{i\varphi}}{2}\, e^{2\pi i n_0 y} + \frac{Ae^{-i\varphi}}{2}\, e^{2\pi i - n_0 y}.$$

Thus the only nonzero Fourier coefficients of h occur at $(m,n) = (0,n_0)$ and $(m,n) = (0,-n_0)$. By the definition of $\Gamma_n(h;z_0)$, only the term $m = 0$ contributes, and since J_0 is even we obtain

$$\Gamma_{n_0}(h;z_0) = \frac{A}{2}e^{i\varphi}J_0(n_0 z_0), \qquad \Gamma_{-n_0}(h;z_0) = \frac{A}{2}e^{-i\varphi}J_0(n_0 z_0).$$

All remaining vertical frequencies vanish identically. □

Theorem 7.2. *Let*

$$T_{A,n_0,\varphi}(x,y,z) = (x+\alpha,\ y+\beta\sin(2\pi x),\\ z + A\cos(2\pi n_0 y + \varphi)).$$

If α *is Diophantine and*

$$J_0(n_0 z_0) = 0,$$

then $T_{A,n_0,\varphi}$ *satisfies the original Cesàro form of Sarnak's conjecture.*

Proof. By Proposition 7.1, the obstruction vector vanishes if and only if $J_0(n_0 z_0) = 0$. The conclusion now follows from Theorem 6.1. □

Corollary 7.3. *Fix a Diophantine* α*, an integer* $n_0 \geq 1$*, and parameters* $A \neq 0$ *and* $\varphi \in \mathbb{T}$*. Then there exist infinitely many real values of* β *for which* $T_{A,n_0,\varphi}$ *satisfies the original Cesàro form of Sarnak's conjecture.*

Proof. Let $(\xi_m)_{m\geq 1}$ be the positive zeros of J_0. Since J_0 has infinitely many positive zeros, this sequence is infinite. For each m, define

$$\beta_m := \frac{\xi_m \sin(\pi\{\alpha\})}{\pi n_0}.$$

Then

$$n_0 z_0 = n_0 \frac{\pi\beta_m}{\sin(\pi\{\alpha\})} = \xi_m,$$

so $J_0(n_0 z_0) = J_0(\xi_m) = 0$. The conclusion follows from Theorem 7.2. □

Corollary 7.4. *Consider*

$$T(x,y,z) = (x+\alpha,\ y+\beta\sin(2\pi x),\ z+\cos(2\pi y)).$$

If α is Diophantine and

$$J_0(z_0) = 0,$$

then T satisfies the original Cesàro form of Sarnak's conjecture.

Proof. Theorem 7.2 applied with $A = 1$, $n_0 = 1$, and $\varphi = 0$ gives the claim. □

8. Obstruction geometry

The exact denominator law of Theorem 4.1 has a second output beyond the Diophantine rigidity route: it compresses the finite-band resonance problem to an explicit finite-dimensional Bessel quotient. The section first records the one-frequency asymptotic input and the resulting compatibility hierarchy, and then applies them to the quotient. The proofs split into three types: shared-profile finiteness from compatibility plus formal phase-purity, anchored rigidity from exact low-order elimination, and generic thinness from placing the compatibility equations on quotient strata.

The obstruction quotient and its kernel. Fix bandwidths (M,N). Let $\mathcal{H}_{M,N}$ be the real vector space of real-valued trigonometric polynomials

$$h(x,y) = u_0(x) + \sum_{n=1}^{N}\big(G_n(x)e^{2\pi iny} + \overline{G_n(x)e^{2\pi iny}}\big),$$

where u_0 is a real trigonometric polynomial of degree at most M and

$$G_n(x) = \sum_{|m|\le M} g_{m,n}e^{2\pi imx}.$$

Define

$$\mathcal{T}_{M,N} := \mathbb{R} \oplus \bigoplus_{n=1}^{N} \operatorname{span}_{\mathbb{R}}\{J_0(n\cdot),\dots,J_M(n\cdot),\ iJ_0(n\cdot),\dots,iJ_M(n\cdot)\}.$$

The exact obstruction law defines the real-linear map

$$\mathcal{O}_{M,N} : \mathcal{H}_{M,N} \to \mathcal{T}_{M,N}, \qquad h \mapsto \big(\Gamma_0(h;\cdot),\Gamma_1(h;\cdot),\dots,\Gamma_N(h;\cdot)\big).$$

Theorem 8.1. *The map $\mathcal{O}_{M,N}$ is surjective. Its kernel is the explicit subspace $\mathcal{K}_{M,N}$ defined by:*

(i) *the zero-frequency part u_0 has zero mean;*
(ii) *for each $1 \le n \le N$, the coefficients*

$$d_{m,n} := g_{m,n}(-i)^m e^{i\pi ma}$$

satisfy

$$d_{0,n} = 0, \qquad d_{-m,n} = -(-1)^m d_{m,n} \quad (1 \le m \le M).$$

Moreover,

$$\dim_{\mathbb{R}} \mathcal{K}_{M,N} = 2M(N+1), \qquad \mathcal{H}_{M,N}/\mathcal{K}_{M,N} \cong \mathcal{T}_{M,N}.$$

Proof. By Theorem 4.1 and remark 4.3, the obstruction splits by vertical frequency. For $n \geq 1$ one has

$$\Gamma_n(h; z_0) = \sum_{|m| \leq M} d_{m,n} J_m(nz_0).$$

Using $J_{-m} = (-1)^m J_m$, this vanishes identically if and only if

$$d_{0,n} = 0, \qquad d_{-m,n} = -(-1)^m d_{m,n} \quad (1 \leq m \leq M).$$

For $n = 0$, Remark 4.3 gives

$$\Gamma_0(h; z_0) = \widehat{u}_0(0),$$

so the zero-frequency obstruction vanishes exactly when u_0 has zero mean. This proves the kernel description.

For the dimension count, the mean-zero u_0 sector contributes $2M$ real parameters, and each positive frequency contributes another $2M$ real parameters, so

$$\dim_{\mathbb{R}} \mathcal{K}_{M,N} = 2M + N \cdot 2M = 2M(N+1).$$

To prove surjectivity, fix $1 \leq n \leq N$ and let

$$\Psi_n(t) = \sum_{m=0}^{M} (a_m + ib_m) J_m(nt)$$

be an arbitrary element of the nth real summand of $\mathcal{T}_{M,N}$. Define

$$d_{m,n} := a_m + ib_m \quad (0 \leq m \leq M), \qquad d_{-m,n} := 0 \quad (1 \leq m \leq M),$$

and then set

$$g_{m,n} := d_{m,n} i^m e^{-i\pi m a} \qquad (|m| \leq M).$$

Let

$$G_n(x) := \sum_{|m| \leq M} g_{m,n} e^{2\pi i m x}, \qquad h_n(x,y) := G_n(x) e^{2\pi i n y} + \overline{G_n(x) e^{2\pi i n y}}.$$

Then $h_n \in \mathcal{H}_{M,N}$ is real-valued by construction. The reality constraint only ties the $-n$ block to the complex conjugate of the n block; it imposes no relation among the coefficients $g_{m,n}$ within the positive n block itself. Hence

$$\Gamma_n(h_n; t) = \sum_{|m| \leq M} d_{m,n} J_m(nt) = \Psi_n(t).$$

So every element of the nth summand is realized by a real-valued cocycle. The zero-frequency summand $\mathbb{R}$ is realized by choosing the mean of u_0 arbitrarily. Therefore $\mathcal{O}_{M,N}$ is surjective, and the quotient description follows. □

One-frequency asymptotics and compatibility. The quotient geometry rests on two elementary asymptotic inputs: a one-frequency dichotomy, and the resulting compatibility relations for any hypothetical common resonant sequence.

Proposition 8.2. *Let*

$$\Gamma(t) = P(t^{-1})J_0(t) + Q(t^{-1})J_1(t), \qquad P, Q \in \mathbb{C}[u],$$

be nontrivial. Write

$$r := \min\{\operatorname{ord}_0 P, \operatorname{ord}_0 Q\}, \quad P(u) = u^r \widetilde{P}(u), \quad Q(u) = u^r \widetilde{Q}(u),$$

and set

$$\alpha := \widetilde{P}(0), \qquad \beta := \widetilde{Q}(0).$$

Then:

(i) *if*

$$\Im(\overline{\alpha}\beta) \neq 0,$$

then Γ *has only finitely many real zeros;*

(ii) *if*

$$\Im(\overline{\alpha}\beta) = 0,$$

then, after factoring out a common complex phase, the positive zeros of Γ *admit an asymptotic expansion*

$$t_k = k\pi + \delta + \sum_{r=1}^{L} \lambda_r (k\pi)^{-r} + O(k^{-L-1})$$

to every finite order L*.*

Proof. Using the standard Hankel expansions of J_0 and J_1 [2, §10.17,§10.20], one has

$$\Gamma(t) = \sqrt{\frac{2}{\pi}}\, t^{-r-\frac{1}{2}} \big(\alpha \cos\xi + \beta \sin\xi + O(t^{-1})\big), \qquad \xi := t - \frac{\pi}{4}.$$

If $\Im(\overline{\alpha}\beta) \neq 0$, then the complex oscillatory profile $\alpha\cos\xi + \beta\sin\xi$ never vanishes on $\mathbb{R}$, hence has a positive lower bound on the circle. Therefore $|\Gamma(t)| \gg t^{-r-1/2}$ for large t, proving *(i)*.

If $\Im(\overline{\alpha}\beta) = 0$, then α and β are real-linearly dependent in $\mathbb{C}$. Factoring out a common complex phase reduces the leading oscillatory profile to a real combination $A\cos\xi + B\sin\xi$. The full Hankel series then gives a formal zero equation with nonvanishing derivative at the leading root. The coefficients of the zero-shift series are then obtained recursively by coefficient matching, equivalently by applying the implicit function theorem to the successive truncated equations; compare the standard asymptotic construction of large Bessel zeros in [2, §10.21]. This yields the expansion in *(ii)*. □

Proposition 8.3. *Let $\mathcal{N}$ be a set of active positive frequencies, and for each $n \in \mathcal{N}$ assume that Γ_n is in the oscillatory-real branch of Proposition 8.2, with positive zeros*

$$t_{n,k} = k\pi + \delta_n + \sum_{r=1}^{L} \lambda_{n,r}(k\pi)^{-r} + O(k^{-L-1}).$$

If there exists an unbounded common real resonant sequence, then for every pair of distinct active frequencies $m, n \in \mathcal{N}$ one has

$$n\delta_m - m\delta_n \in \pi\mathbb{Z}$$

and

$$m^{r+1}\lambda_{n,r} = n^{r+1}\lambda_{m,r} \qquad (1 \le r \le L).$$

Proof. Let $z_j \to \infty$ be the common base parameters, so that nz_j and mz_j lie on the respective zero sequences. Thus

$$nz_j = k_{n,j}\pi + \delta_n + \sum_{r=1}^{L} \lambda_{n,r}(k_{n,j}\pi)^{-r} + O(z_j^{-L-1}),$$

$$mz_j = k_{m,j}\pi + \delta_m + \sum_{r=1}^{L} \lambda_{m,r}(k_{m,j}\pi)^{-r} + O(z_j^{-L-1}).$$

Multiply the first identity by m and the second by n, then subtract. Since

$$k_{n,j} = \frac{n}{\pi}z_j + O(1), \qquad k_{m,j} = \frac{m}{\pi}z_j + O(1),$$

comparison of the constant term gives

$$n\delta_m - m\delta_n \in \pi\mathbb{Z}.$$

Once this is fixed, comparing the successive powers of z_j^{-1} yields the higher-order relations inductively. □

Proposition 8.4. *Let*

$$F_b(t) = \sum_{m\in S} c_m J_m(t)$$

be a nontrivial reduced real Bessel profile in the oscillatory-real branch, and let

$$\mathcal{G}_b(u) := \sum_{m\in S} c_m(-i)^m H_m(u)$$

be its formal Hankel amplitude. If all higher zero-shift coefficients of F_b vanish, then there exists $\phi \in \mathbb{R}$ such that

$$e^{-i\phi}\mathcal{G}_b(u) \in \mathbb{R}[[u]].$$

Proof. Choose the leading phase ϕ so that

$$e^{-i\phi}\mathcal{G}_b(u) = \mathcal{A}(u) - i\mathcal{B}(u),$$

with $\mathcal{A}(0) > 0$ and $\mathcal{B}(0) = 0$. The formal zero equation near the positive oscillatory roots is

$$0 = -\mathcal{A}(u)\sin s + \mathcal{B}(u)\cos s,$$

where $s = s(u) \in u\mathbb{R}[[u]]$ is the formal zero-shift series. If all higher zero-shift coefficients vanish, then $s(u) \equiv 0$, so the equation reduces to $\mathcal{B}(u) \equiv 0$. Therefore $e^{-i\phi}\mathcal{G}_b(u) = \mathcal{A}(u)$ is real as a formal series. □

Shared-profile multi-frequency finiteness. We next isolate a large class in which multi-frequency infinite resonance is impossible. This is the first genuinely global finiteness statement beyond the one-frequency sector.

Theorem 8.5. *Let*

$$F_b(t) = \sum_{m \in S} c_m J_m(t)$$

be a nontrivial reduced real Bessel profile with finite support $S \subset \mathbb{Z}_{\geq 0}$. Assume that at least two distinct positive active vertical frequencies carry this same reduced profile, up to nonzero real scalar multiples. Then the common real resonant parameter set is finite.

Proof. Assume for contradiction that the common real resonant parameter set is infinite. Since the active frequencies carry the same reduced profile, all of them have the same one-frequency asymptotic coefficients $\delta, \lambda_1, \lambda_2, \ldots$. Applying Proposition 8.3 to two distinct active frequencies shows that every higher coefficient must vanish:

$$\lambda_r = 0 \qquad (r \geq 1).$$

By Proposition 8.4, there exists $\phi \in \mathbb{R}$ such that

$$\mathcal{G}_b(u) := \sum_{m \in S} c_m (-i)^m H_m(u)$$

has the property that

$$e^{-i\phi}\mathcal{G}_b(u) \in \mathbb{R}[[u]].$$

Write

$$H_m(u) = E_m(u) + i\, O_m(u),$$

where E_m is even and O_m is odd in u. The imaginary part of $e^{-i\phi}\mathcal{G}_b(u)$ therefore yields

$$\sum_{m \in S} c_m \sin(\phi + m\pi/2)\, E_m(u) = 0,$$

$$\sum_{m \in S} c_m \cos(\phi + m\pi/2)\, O_m(u) = 0.$$

More concretely, the coefficient of u^{2r} in E_m has the form

$$\kappa_r m^{4r} + (\text{lower powers of } m^2), \qquad \kappa_r \neq 0,$$

while the coefficient of u^{2r+1} in O_m has the form

$$\kappa'_r m^{4r+2} + (\text{lower powers of } m^2), \qquad \kappa'_r \neq 0.$$

Choose distinct indices $m_1, \dots, m_s \in S$. Evaluating the coefficients of $u^0, u^2, \dots, u^{2s-2}$ in a linear relation among the E_{m_j} gives a square matrix whose leading part is the Vandermonde matrix

$$\Big((m_j^2)^r\Big)_{\substack{0\le r\le s-1\\ 1\le j\le s}},$$

which is invertible because the numbers m_j^2 are distinct. Hence $\{E_m\}_{m\in S}$ is linearly independent. The same argument, now using the coefficients of $u^1, u^3, \dots, u^{2s-1}$, shows that $\{O_m\}_{m\in S}$ is linearly independent as well.

Therefore every active $m \in S$ satisfies

$$c_m \sin(\phi + m\pi/2) = 0, \qquad c_m \cos(\phi + m\pi/2) = 0,$$

so $c_m = 0$ for every $m \in S$. This contradicts the nontriviality of F_b. □

J_0-anchored support-two rigidity. The shared-profile theorem closes the synchronized multi-frequency sector. We now record a different geometric phenomenon: once one active frequency is the pure vertical anchor J_0, the entire support-two non-shared sector is already closed.

Theorem 8.6. *Assume that one active positive vertical frequency carries the pure vertical anchor profile J_0. Let $k \ge 1$, and let a second active positive vertical frequency carry a nontrivial profile supported on $\{0, k\}$:*

$$F(t) = b_0 J_0(t) + b_k J_k(t), \qquad (b_0, b_k) \neq (0, 0).$$

If the two active frequencies belong to a common infinite-resonance configuration, then

$$b_k = 0.$$

Equivalently, the second profile must again be a scalar multiple of J_0.

Proof. Write the two active frequencies as n and p, and let

$$q := \frac{p^2}{n^2} > 0.$$

If $b_0 = 0$, then the second profile is a pure Bessel mode J_k. For even k the first zero-shift coefficient is negative, while $\lambda_1(J_0) = 1/8 > 0$, so normalized equality at order 1 is impossible. For odd k one has

$$\lambda_1(J_k) = \frac{4k^2 - 1}{8},$$

so normalized equality would force

$$\frac{p^2}{n^2} = 4k^2 - 1.$$

But

$$(2k-1)^2 < 4k^2 - 1 < (2k)^2,$$

so $4k^2 - 1$ is not a square; this is impossible because p^2/n^2 is a rational square.

It remains to treat the mixed case $b_0 \neq 0$.

If k is odd, write $k = 2r + 1$,

$$a := 4k^2 - 1, \qquad y := (-1)^r \frac{b_k}{b_0}.$$

A direct Hankel expansion gives

$$\lambda_1 = \frac{1 - ay^2}{8(1 + y^2)}, \qquad \lambda_2 = \frac{y(a+1)\big((a+9)y^2 - (a-7)\big)}{128(1+y^2)^2}.$$

Compatibility with the anchor J_0 forces $\lambda_2 = 0$. Hence either $y = 0$, which yields $b_k = 0$, or

$$y^2 = \frac{a-7}{a+9}.$$

In the latter case normalized equality at order 1 gives

$$q = 8\lambda_1 = \frac{1 - ay^2}{1 + y^2} = -\frac{(a-9)(a-1)}{2(a+1)} < 0,$$

contradiction.

If k is even, write $k = 2r$,

$$a := 4k^2 - 1, \qquad y := (-1)^r \frac{b_k}{b_0}.$$

Then the same-parity Hankel expansion yields

$$\lambda_1 = \frac{1 - ay}{8(1 + y)}.$$

Compatibility with the anchor implies, after eliminating the third-order coefficient,

$$y\big(A(a) + B(a)y + C(a)y^2\big) = 0,$$

where

$$A(a) = a^3 - 29a^2 - 253a - 223,$$
$$B(a) = -a^3 + 157a^2 - 67a - 225,$$
$$C(a) = 192a(a+1).$$

If $y \neq 0$, then using

$$q = \frac{1 - ay}{1 + y}$$

to eliminate y gives

$$\begin{aligned} 2(a+1)^3 q^2 + \big(3a^4 - 216a^3 - 666a^2 - 672a - 225\big)q \\ + a\big(a^4 - 30a^3 - 96a^2 - 98a - 33\big) = 0. \end{aligned}$$

For $k \geq 4$ one has $a \geq 63$, and all three coefficients above are strictly positive, which is impossible for $q > 0$. The remaining case $k = 2$ is an explicit low-order calculation of the same type and again forces $b_2 = 0$.

Thus in every case $b_k = 0$. □

A second-order normal form in the anchored even-odd support-three sector.

Proposition 8.7. *Let $m \geq 2$ be even and $n \geq 1$ odd. Set*

$$\widetilde{J}_m := (-1)^{m/2} J_m, \qquad \widetilde{J}_n := (-1)^{(n-1)/2} J_n, \qquad M := m^2, \qquad N := n^2.$$

Consider the reduced profile

$$F(t) = J_0(t) + b\widetilde{J}_m(t) + c\widetilde{J}_n(t).$$

Then the first two one-frequency shift coefficients of F are

$$\lambda_1(F) = -\frac{M\, b(b+1) + (N-1)c^2}{2\big((1+b)^2 + c^2\big)},$$

and

$$\lambda_2(F) = -\frac{c\big((b+1)\mathcal{Q}_{M,N}(b) + c^2\mathcal{R}_{M,N}(b)\big)}{8\big((1+b)^2 + c^2\big)^2},$$

where

$$\begin{aligned}\mathcal{Q}_{M,N}(b) = N^2 - 2N + 1 \\ &+ b(-M^2 - 2MN + 6M + 2N^2 - 4N + 2) \\ &+ b^2(M^2 - 2MN + 6M + N^2 - 2N + 1),\end{aligned}$$

$$\mathcal{R}_{M,N}(b) = -M^2 b + 2MNb + 2Mb - N^2 b - N^2 + 2Nb + 2N - b - 1.$$

Proof. Using the exact Bessel recurrence in the (J_0, J_1) basis, one has

$$\widetilde{J}_m(t) = \Big(1 - \frac{M(M-4)}{8}u^2 + O(u^4)\Big)J_0(t) \;+\; \Big(-\frac{M}{2}u + O(u^3)\Big)J_1(t),$$

$$\begin{aligned}\widetilde{J}_n(t) = \Big(\frac{N-1}{2}u + O(u^3)\Big)J_0(t) \qquad\qquad u = t^{-1}. \\ + \Big(1 - \frac{(N-1)^2}{8}u^2 + O(u^4)\Big)J_1(t),\end{aligned}$$

Hence

$$F(t) = A(u)J_0(t) + B(u)J_1(t),$$

with

$$A(u) = (1+b) + \frac{N-1}{2}cu - \frac{M(M-4)}{8}bu^2 + O(u^3),$$

$$B(u) = c - \frac{M}{2}bu - \frac{(N-1)^2}{8}cu^2 + O(u^3).$$

Rotating the leading phase so that the tangent series has vanishing constant term and expanding that tangent series to order u^2 gives the displayed formulas for $\lambda_1(F)$ and $\lambda_2(F)$. □

Corollary 8.8. *Retain the notation of Proposition 8.7, and assume*

$$c \neq 0, \qquad b \neq -1, \qquad \lambda_2(F) = 0.$$

Then:

(i) *the genuinely support-three branch is contained in the explicit candidate curve*
$$c^2 = -\frac{(b+1)\mathcal{Q}_{M,N}(b)}{\mathcal{R}_{M,N}(b)};$$
(ii) *along that candidate curve,* λ_1 *collapses to the fractional-linear function*
$$\lambda_1 = \frac{-M^2b + 4Mb + N^2b + N^2 - 2Nb - 2N + b + 1}{4(Mb - Nb - N + b + 1)};$$
(iii) *its derivative is*
$$\frac{d\lambda_1}{db} = \frac{M(N-1)(M-N-3)}{4(Mb - Nb - N + b + 1)^2}.$$

In particular, for genuine even-odd support-three data one has $M-N-3 \neq 0$*, so* λ_1 *is a strict coordinate on every real branch of the second-order candidate curve.*

Proof. Part *(i)* is the equation $\lambda_2(F) = 0$ from Proposition 8.7. Substituting that relation into the formula for $\lambda_1(F)$ yields part *(ii)*, and differentiating the resulting fractional-linear expression gives part *(iii)*. Finally, if $M-N-3 = 0$, then
$$(2k)^2 - (2r+1)^2 = 3,$$
which is impossible because it would factor as
$$\big(2k - (2r+1)\big)\big(2k + (2r+1)\big) = 3$$
with no integer solutions. □

Second-order projective rigidity in the anchored even-odd sector.

Theorem 8.9. *Assume that one active positive vertical frequency carries the pure vertical anchor profile* J_0*. Let a second active positive vertical frequency carry a profile of the form*
$$F(t) = aJ_0(t) + bJ_m(t) + cJ_n(t), \qquad a \neq 0, \qquad m \geq 2 \text{ even}, \qquad n \geq 1 \text{ odd}.$$
If the two active frequencies belong to a common infinite-resonance configuration, then
$$c = 0.$$
Equivalently, the anchored even-odd support-three sector is disjoint from the J_0 *fiber of the normalized Prüfer map already at second order.*

Proof. By scaling, we may assume $a = 1$ and replace J_m, J_n by the normalized modes
$$\widetilde{J}_m := (-1)^{m/2} J_m, \qquad \widetilde{J}_n := (-1)^{(n-1)/2} J_n,$$
which does not affect real solvability. Retain the notation
$$M := m^2, \qquad N := n^2.$$
Suppose first that $c \neq 0$.

If $b = -1$, then Proposition 8.7 gives

$$\lambda_1(F) = \frac{1-N}{2} < 0,$$

which is incompatible with the anchor normalization

$$q := 8\lambda_1(F) > 0.$$

So we may also assume $b \neq -1$.

By Corollary 8.8, the conditions

$$c \neq 0, \qquad b \neq -1, \qquad \lambda_2(F) = 0$$

force F to lie on the second-order candidate curve, and along that curve the relation

$$q = 8\lambda_1(F) > 0$$

has the unique solution

$$b = \frac{2N^2 + Nq - 4N - q + 2}{2M^2 + Mq - 8M - 2N^2 - Nq + 4N + q - 2}.$$

Substituting this into the candidate-curve equation from Corollary 8.8*(i)* yields

$$c^2 = -\frac{M^2(2M+q-8)\big(q^2 + (2M+4N-12)q + 8(N-1)^2\big)}{(4N+q-4)\big(2M^2 + Mq - 8M - 2N^2 - Nq + 4N + q - 2\big)^2}.$$

Now

$$4N + q - 4 > 0, \qquad 2M + q - 8 \geq q > 0,$$

and

$$q^2 + (2M + 4N - 12)q + 8(N-1)^2 > 0.$$

Hence the right-hand side is strictly negative, contradicting real-valuedness of c^2.

Therefore $c \neq 0$ is impossible, and so $c = 0$. □

Second-order projective rigidity in the anchored odd-odd sector.

Theorem 8.10. *Assume that one active positive vertical frequency carries the pure vertical anchor profile J_0. Let a second active positive vertical frequency carry a profile of the form*

$$F(t) = aJ_0(t) + bJ_m(t) + cJ_n(t), \qquad a \neq 0, \qquad m, n \geq 1 \text{ odd}, \qquad m \neq n.$$

If the two active frequencies belong to a common infinite-resonance configuration, then

$$b = c = 0.$$

Proof. By scaling, we may assume $a = 1$. Normalize the odd modes by

$$\widetilde{J}_m := (-1)^{(m-1)/2} J_m, \qquad \widetilde{J}_n := (-1)^{(n-1)/2} J_n,$$

and write

$$M := m^2, \qquad N := n^2.$$

Set

$$u := b + c, \qquad d := (M-1)b + (N-1)c.$$

In these coordinates the first two normalized Prüfer coefficients are

$$\lambda_1(F) = -\frac{du}{2(1+u^2)},$$

$$\lambda_2(F) = \frac{2ud^2 - (M+N-2)(1+u^2)d + (M-1)(N-1)u(1+u^2)}{8(1+u^2)^2}.$$

Now impose the anchor normalization

$$q := 8\lambda_1(F) > 0.$$

Then

$$d = -\frac{q(1+u^2)}{4u},$$

and substituting this into $\lambda_2(F) = 0$ gives

$$u^2 = -\frac{q\big(q + 2(M+N-2)\big)}{q^2 + 2(M+N-2)q + 8(M-1)(N-1)}.$$

The right-hand side is strictly negative for every $q > 0$, which is impossible over the reals. Hence no genuinely support-three anchored odd-odd configuration exists. The remaining support-two cases are excluded by Theorem 8.6, so necessarily $b = c = 0$. □

The anchored even-even sector.

Proposition 8.11. *Let*

$$F(t) = J_0(t) + b\widetilde{J}_m(t) + c\widetilde{J}_n(t), \qquad m, n \geq 2 \text{ even}, \qquad m \neq n,$$

where

$$\widetilde{J}_\ell := (-1)^{\ell/2} J_\ell, \qquad M := m^2, \qquad N := n^2.$$

Set

$$U := b + c, \qquad D := Mb + Nc.$$

Then

$$\lambda_1(F) = -\frac{D}{2(U+1)}, \qquad \lambda_2(F) = 0.$$

If moreover

$$q := 8\lambda_1(F) > 0, \qquad \lambda_3(F) = -\frac{25}{384}q^2,$$

then the even-even support-three branch is reduced to the unique candidate curve

$$U(q) = -q\,\frac{\mathcal{N}_{M,N}(q)}{\mathcal{D}_{M,N}(q)},$$

where

$$\begin{aligned}\mathcal{N}_{M,N}(q) = {} & 8M^2 + 8MN + 6Mq - 64M \\ & + 8N^2 + 6Nq - 64N + q^2 - 124q + 128,\end{aligned}$$

$$\begin{aligned}\mathcal{D}_{M,N}(q) = 32M^2N + 8M^2q + 32MN^2 + 32MNq - 256MN \\ + 6Mq^2 - 64Mq + 8N^2q + 6Nq^2 \\ - 64Nq + q^3 - 124q^2 + 128q,\end{aligned}$$

and along that curve one has

$$\lambda_5(F) - \frac{1073}{5120}q^3 = \frac{-q\,\mathcal{P}_{M,N}(q)}{491520\,(4M + 4N + 3q - 32)},$$

where $\mathcal{P}_{M,N}(q)$ *is the explicit symmetric quintic*

$$\mathcal{P}_{M,N}(q) = q^5 + (8S - 564)q^4 + A_{M,N}q^3 + B_{M,N}q^2 + C_{M,N}q + D_{M,N},$$

with

$$S := M + N, \qquad P := MN.$$

$$\begin{aligned}A_{M,N} &= 20S^2 + 20P - 2640S + 278544,\\ B_{M,N} &= 16S^3 + 88PS - 3120S^2 - 3760P + 496000S - 3727616,\\ C_{M,N} &= 64P^2 + 96PS^2 - 1152PS - 23552P\\ &\quad - 1984S^3 + 81920S^2 - 1068032S + 4726784,\\ D_{M,N} &= 128P^2S - 4096P^2 - 1024PS^2 + 36864PS\\ &\quad - 131072P + 2048S^3 - 81920S^2 + 557056S - 1048576.\end{aligned}$$

Proof. For even normalized modes one has

$$\begin{aligned}\widetilde{J}_\ell(t) = &\Big(1 - \frac{\ell^2(\ell^2-4)}{8}u^2 + \frac{\ell^2(\ell^2-4)^2(\ell^2-16)}{384}u^4 + O(u^6)\Big)J_0(t)\\ &+ \Big(-\frac{\ell^2}{2}u + \frac{\ell^2(\ell^2-4)^2}{48}u^3\\ &\quad - \frac{\ell^2(\ell^2-4)^2(\ell^2-16)^2}{3840}u^5 + O(u^7)\Big)J_1(t),\end{aligned}$$

Here $u = t^{-1}$. Substituting these expansions into

$$F(t) = A(u)J_0(t) + B(u)J_1(t)$$

and then into the general odd-order Prüfer formulas yields

$$\lambda_1(F) = -\frac{D}{2(U+1)}, \qquad \lambda_2(F) = 0.$$

Imposing

$$q = 8\lambda_1(F) > 0$$

gives

$$D = -\frac{q(U+1)}{4}.$$

Substituting this into the order-3 relation

$$\lambda_3(F) = -\frac{25}{384}q^2$$

produces a linear equation in U, whose unique solution is the displayed rational function $U(q)$. Finally, substituting both

$$D = -\frac{q(U+1)}{4}, \qquad U = U(q)$$

into the order-5 coefficient gives the displayed residual formula with the explicit symmetric quintic $\mathcal{P}_{M,N}(q)$. □

Theorem 8.12. *Assume that one active positive vertical frequency carries the pure vertical anchor profile J_0. Let a second active positive vertical frequency carry a profile of the form*

$$F(t) = aJ_0(t) + bJ_m(t) + cJ_n(t), \qquad a \neq 0, \qquad m, n \geq 2 \text{ even}, \qquad m \neq n.$$

If the two active frequencies belong to a common infinite-resonance configuration, then

$$b = c = 0.$$

Proof. By scaling, we may assume $a = 1$, and we replace J_m, J_n by the normalized even modes

$$\widetilde{J}_m := (-1)^{m/2} J_m, \qquad \widetilde{J}_n := (-1)^{n/2} J_n.$$

Suppose first that

$$bc \neq 0.$$

Then we are in the genuinely support-three even-even sector, so Proposition 8.11 applies. Write

$$M = 4x^2, \qquad N = 4y^2, \qquad 2 \leq x < y,$$

and let $\mathcal{P}_{x,y}(q) := \mathcal{P}_{M,N}(q)$.

We claim that

$$\mathcal{P}_{x,y}(q) > 0 \qquad (q > 0).$$

If $x \geq 3$, then

$$\mathcal{P}_{x,y}(0) = 131072(x-1)^2(x+1)^2(y-1)^2(y+1)^2(x^2+y^2-8) > 0,$$

so it is enough to prove

$$\mathcal{P}'_{x,y}(q) > 0 \qquad (q \geq 0).$$

Write

$$a := x^2, \qquad d := y^2 - x^2 > 0.$$

Then

$$\begin{aligned}\mathcal{P}'_{x,y}(q) = 5q^4 &+ (128x^2 + 128y^2 - 2128)q^3 \\ &+ 48Q_2(a,d)q^2 + 512Q_1(a,d)q + 4096Q_0(a,d),\end{aligned}$$

where

$$\begin{aligned}Q_2(a,d) &= 20a^2 + 20ad - 264a + 4d^2 - 132d + 17409, \\ Q_1(a,d) &= 76a^3 + 114a^2d - 1015a^2 + 46ad^2 - 1015ad + 15500a \\ &\quad + 4d^3 - 195d^2 + 7750d - 14561,\end{aligned}$$

$$\begin{aligned}Q_0(a,d) &= 28a^4 + 56a^3d - 284a^3 + 34a^2d^2 - 426a^2d + 1188a^2 \\ &\quad + 6ad^3 - 204ad^2 + 1188ad - 2086a \\ &\quad - 31d^3 + 320d^2 - 1043d + 1154.\end{aligned}$$

Since $d \geq 2x+1$ and

$$\partial_d Q_2(a,d) = 4(5a + 2d - 33) > 0,$$

$$\partial_d Q_1(a,d) = 114a^2 + 92ad - 1015a + 12d^2 - 390d + 7750 > 0,$$

$$\begin{aligned}\partial_d Q_0(a,d) &= 56a^3 + 68a^2d - 426a^2 + 18ad^2 - 408ad + 1188a \\ &\quad - 93d^2 + 640d - 1043 > 0.\end{aligned}$$

throughout the region $x \geq 3$, $d \geq 2x+1$, the coefficients are minimized by the adjacent pair $y = x+1$. It is therefore enough to check

$$\Pi_x(q) := \mathcal{P}'_{x,x+1}(q) > 0.$$

For every $x \geq 3$,

$$\Pi_x(q) = 5q^4 + \alpha_x q^3 + \beta_x q^2 + \gamma_x q + \delta_x,$$

$$\begin{aligned}\alpha_x &= 256x^2 + 256x - 2128, \\ \beta_x &= 48(100x^4 + 200x^3 - 1140x^2 - 1240x + 16769), \\ \gamma_x &= 512(76x^6 + 228x^5 - 717x^4 - 1814x^3 + 13799x^2 + 14744x - 7002), \\ \delta_x &= 16384(x-1)^2(x+2)^2(7x^4 + 14x^3 - 30x^2 - 37x + 25).\end{aligned}$$

and every coefficient is strictly positive. Hence $\Pi_x(q) > 0$ for $q \geq 0$, so

$$\mathcal{P}_{x,y}(q) > 0 \qquad (q > 0)$$

throughout the large-even interior.

If $x = 2$ and $y \geq 5$, write

$$r := y^2 - 9 \geq 16, \qquad q = (r+8)z.$$

After dividing by $(r+8)^2$, the residual becomes

$$\begin{aligned}G_r(z) &= r^3z^5 + 32r^3z^4 + 320r^3z^3 + 1024r^3z^2 \\ &\quad + 24r^2z^5 + 364r^2z^4 + 1600r^2z^3 \\ &\quad + 12544r^2z^2 - 28672r^2z \\ &\quad + 192rz^5 - 320rz^4 + 199184rz^3 \\ &\quad + 1460224rz^2 - 4096rz + 1179648r \\ &\quad + 512z^5 - 9472z^4 + 1654912z^3 \\ &\quad + 16347648z^2 + 1802240z + 5898240.\end{aligned}$$

For $r \geq 16$ the coefficients of z^5, z^4, and z^3 are strictly positive, and one can decompose

$$G_r(z) = z^3(a_5z^2 + a_4z + a_3) + (a_2z^2 - bz + c),$$

with all of a_5, a_4, a_3 positive, and

$$a_2 z^2 - bz + c > 0 \qquad (z > 0),$$

because

$$a_2 = 1024r^3 + 12544r^2 + 1460224r + 16347648,$$
$$b = 28672r^2 + 4096r - 1802240, \qquad c = 1179648r + 5898240,$$
$$4a_2c - b^2 = 16777216(239r^4 + 4954r^3 + 434487r^2 + 6652096r + 22795280) > 0.$$

Hence $G_r(z) > 0$ for all $z > 0$, so $\mathcal{P}_{16,N}(q) > 0$ for every $q > 0$ whenever $N \geq 100$.

It remains to treat the isolated pairs $(x, y) = (2, 3)$ and $(2, 4)$, i.e.

$$(M, N) = (16, 36) \qquad \text{and} \qquad (M, N) = (16, 64).$$

Here

$$\mathcal{P}_{16,36}(q) = q^5 - 148q^4 + 206864q^3 + 16347648q^2 + 14417920q + 377487360,$$
$$\mathcal{P}_{16,64}(q) = q^5 + 76q^4 + 215824q^3 + 27535104q^2 + 5529600q + 3185049600.$$

The second polynomial is manifestly positive on $(0, \infty)$. For the first one, a direct Sturm count shows that it has no positive real root. Since its constant term is positive, it also satisfies

$$\mathcal{P}_{16,36}(q) > 0 \qquad (q > 0).$$

Thus $\mathcal{P}_{M,N}(q) > 0$ for every pair of distinct even squares M, N and every $q > 0$. This contradicts the fifth-order anchored compatibility equation from Proposition 8.11. Therefore the case $bc \neq 0$ is impossible.

The remaining support-two cases are excluded by Theorem 8.6. Hence

$$b = c = 0.$$

□

Generic finite resonance on the quotient. We finally return to the general multi-frequency quotient geometry. Let $\mathcal{N}$ be a set of active positive frequencies with $|\mathcal{N}| \geq 2$. On a fixed quotient stratum, each active frequency has a one-frequency reduction

$$\Gamma_n(t) = P_n(t^{-1})J_0(nt) + Q_n(t^{-1})J_1(nt),$$

with complex polynomials P_n, Q_n, and therefore a leading oscillatory pair $(\alpha_n, \beta_n) \in \mathbb{C}^2 \setminus \{(0, 0)\}$. By Proposition 8.2, the necessary condition for infinitely many real zeros is

$$R_n := \Im(\overline{\alpha_n}\beta_n) = 0.$$

On the corresponding oscillatory-real chart, the positive zeros admit asymptotic expansions

$$t_{n,k} = k\pi + \delta_n + \sum_{r=1}^{L} \lambda_{n,r}(k\pi)^{-r} + O(k^{-L-1}),$$

and Proposition 8.3 forces all normalized coefficients to agree.

Theorem 8.13. *Fix a quotient stratum with at least two active positive frequencies. Then the subset of points admitting infinitely many common real resonant parameters is contained in a countable union of proper real-analytic subsets. In particular, on each such stratum the infinite-resonance locus is meagre and has Lebesgue measure zero.*

Proof. If an active frequency n satisfies $R_n \neq 0$, then Proposition 8.2 shows that Γ_n has only finitely many real zeros. Hence infinite common resonance is possible only on the oscillatory-real chart defined by

$$R_n = 0 \qquad (n \in \mathcal{N}).$$

Fix a base frequency $n_1 \in \mathcal{N}$. For each integer vector

$$\mathbf{q} = (q_n)_{n \in \mathcal{N} \setminus \{n_1\}} \in \mathbb{Z}^{|\mathcal{N}|-1}$$

and each order $L \geq 1$, let $\mathscr{B}_{L,\mathbf{q}}$ be the real-analytic subset cut out by

$$n\delta_{n_1} - n_1\delta_n = q_n\pi \qquad (n \neq n_1),$$

and

$$n_1^{r+1}\lambda_{n,r} = n^{r+1}\lambda_{n_1,r} \qquad (n \neq n_1,\ 1 \leq r \leq L).$$

By Proposition 8.3, every point with infinitely many common real resonant parameters lies in

$$\bigcup_{\mathbf{q}} \mathscr{B}_{L,\mathbf{q}}$$

for every L.

It remains to show properness. This already follows for $L = 1$. On a fixed quotient stratum, each function R_n depends only on the corresponding one-frequency factor, and the projection onto that factor is surjective. On that factor there are explicit obstruction data with $R_n \neq 0$, for example the profile $J_0 + iJ_1$. Hence R_n is not identically zero on the stratum, so the oscillatory-real equation $R_n = 0$ cuts out a proper real-analytic subset. Therefore each $\mathscr{B}_{1,\mathbf{q}}$ is proper, and the infinite-resonance locus is contained in a countable union of proper real-analytic subsets.

The meagreness and measure-zero statements are standard consequences, since a proper real-analytic subset of a finite-dimensional real-analytic manifold has empty interior and Lebesgue measure zero. □

Corollary 8.14. *Let $\Xi \in \mathcal{T}_{M,N}$ be a quotient datum.*

(i) *If all positive-frequency obstruction components of Ξ vanish identically, then Ξ lies in the beta-uniform obstruction-free sector.*
(ii) *If exactly one positive frequency is active, then the real resonant parameter set is governed by the one-frequency dichotomy of Proposition 8.2: it is finite in the asymptotically nonresonant branch and infinite in the oscillatory-real branch.*
(iii) *If at least two distinct positive active frequencies carry the same reduced profile, then the real resonant parameter set is finite.*

(iv) *If one active positive frequency is the anchor J_0 and another is supported on $\{0, k\}$ for some $k \geq 1$, then the real resonant parameter set is finite.*

(v) *If one active positive frequency is the anchor J_0 and another lies in the anchored support-three sector, then the real resonant parameter set is finite. More precisely: the even-odd and odd-odd sectors are excluded at second order, and the even-even sector is excluded at fifth order.*

(vi) *If at least two positive active frequencies are present in a multi-frequency quotient stratum, then outside a countable union of proper real-analytic subsets of that stratum the real resonant parameter set is finite.*

Proof. Part *(i)* is exactly the kernel description in Theorem 8.1. Part *(ii)* is Proposition 8.2. Part *(iii)* is Theorem 8.5, part *(iv)* is Theorem 8.6, part *(v)* is the combination of Theorems 8.9, 8.10 and 8.12, and part *(vi)* is Theorem 8.13. □

Remark 8.15. Corollary 8.14 describes the current boundary of the problem inside the finite-dimensional obstruction quotient. The unresolved sector is no longer general multi-frequency resonance. It is the genuinely non-synchronized, oscillatory-real, effective multi-frequency sector that survives the explicit kernel, the shared-profile theorem, and the anchored support-two and support-three rigidity results. In particular, once a J_0 anchor is present, the entire support-three picture is now closed.

9. Finite-Jet and Phase Geometry on Bounded Bandwidth Windows

The support-two and support-three results of Section 8 are low-order shadows of a more systematic bounded-bandwidth phase geometry. In this section we isolate the finite-jet and phase-geometric mechanism behind that picture.

A central-factorial basis for the normalized Lommel coefficients. For even and odd normalized modes, write

$$\widetilde{J}_{2k}(t) = \widetilde{A}_{2k}(u)J_0(t) + \widetilde{B}_{2k}(u)J_1(t),$$
$$\widetilde{J}_{2k+1}(t) = \widetilde{A}_{2k+1}(u)J_0(t) + \widetilde{B}_{2k+1}(u)J_1(t), \qquad u = t^{-1},$$

with

$$\widetilde{J}_{2k} := (-1)^k J_{2k}, \qquad \widetilde{J}_{2k+1} := (-1)^k J_{2k+1}.$$

These coefficients are governed by the exact Bessel recurrence in the (J_0, J_1) basis, or equivalently by the classical Lommel polynomials; see [2, §10.74].

Theorem 9.1. *For every $k \geq 1$,*

$$\widetilde{A}_{2k}(u) = \sum_{j=0}^{k-1} E_j((2k)^2)u^{2j}, \qquad \widetilde{B}_{2k}(u) = \sum_{j=0}^{k-1} F_j((2k)^2)u^{2j+1},$$

and for every $k \geq 0$,

$$\widetilde{A}_{2k+1}(u) = \sum_{j=0}^{k} G_j\big((2k+1)^2\big)u^{2j+1}, \qquad \widetilde{B}_{2k+1}(u) = \sum_{j=0}^{k} H_j\big((2k+1)^2\big)u^{2j}.$$

The coefficient polynomials are given by

$$E_0(M) = 1,$$

$$E_j(M) = (-1)^j \, \frac{M(M-4)^2(M-16)^2\cdots(M-(2j-2)^2)^2(M-(2j)^2)}{4^j(2j)!}$$

for $j \geq 1$,

$$F_j(M) = (-1)^{j+1} \, \frac{M(M-4)^2(M-16)^2\cdots(M-(2j)^2)^2}{2\cdot 4^j(2j+1)!},$$

$$G_j(N) = (-1)^j \, \frac{(N-(2j+1)^2)(N-1)^2(N-9)^2\cdots(N-(2j-1)^2)^2}{2\cdot 4^j(2j+1)!},$$

$$H_0(N) = 1, \qquad H_j(N) = (-1)^j \, \frac{(N-1)^2(N-9)^2\cdots(N-(2j-1)^2)^2}{4^j(2j)!}$$

for $j \geq 1$.

In particular, the families E_j, F_j, G_j, H_j form a parity-adapted central-factorial basis in the squared order: they have strict degree growth and vanish at lower same-parity squares with the displayed multiplicities.

Proof. The Lommel representation expresses J_n exactly as a combination of J_0 and J_1 with polynomial coefficients in $u = t^{-1}$. Substituting $n = 2k$ and $n = 2k+1$, inserting the sign normalization, and re-indexing the explicit Lommel sums yields the displayed formulas. The degree and vanishing statements are immediate from the product structure. □

Bounded-bandwidth reconstruction. Fix integers $K \geq 1$ and $L \geq 0$, and consider the bounded even/odd window

$$\mathcal{V}_{K,L} := \left\{ F(t) = a_0 J_0(t) + \sum_{k=1}^{K} b_k \widetilde{J}_{2k}(t) + \sum_{\ell=0}^{L} c_\ell \widetilde{J}_{2\ell+1}(t) \right\}.$$

Write

$$F(t) = A(u)J_0(t) + B(u)J_1(t).$$

By parity,

$$B(u) = \sum_{j=0}^{K-1} \beta_j^{\mathrm{ev}} u^{2j+1} + \sum_{j=0}^{L} \beta_j^{\mathrm{odd}} u^{2j}.$$

The coefficient of u^{2K-1} in the even block comes only from $\widetilde{J}_{2K}$, while the coefficient of u^{2L} in the odd block comes only from $\widetilde{J}_{2L+1}$. This yields a global triangular reconstruction.

Theorem 9.2. *For every bandwidth window (K, L), the coefficient map*

$$\Phi_{K,L} : \mathcal{V}_{K,L} \to \mathbb{R}^{K+L+2},$$

$$F \longmapsto \Big(a_0,\ [u^0]B,\ [u^2]B, \dots, [u^{2L}]B,\ [u^1]B,\ [u^3]B, \dots, [u^{2K-1}]B\Big)$$

is a linear isomorphism.

Equivalently, the full coefficient vector

$$(a_0, b_1, \dots, b_K, c_0, \dots, c_L)$$

is recovered uniquely from a finite jet of the J_1-channel alone.

Proof. By Theorem 9.1, the even contribution to $B(u)$ is triangular in the support index: the coefficient of u^{2K-1} is exactly $b_K F_{K-1}((2K)^2)$ and this scalar is nonzero. Hence b_K is recovered first, then recursively $b_{K-1}, \dots, b_1$. The same argument applies to the odd block, where c_L is recovered from the coefficient of u^{2L}, then recursively $c_{L-1}, \dots, c_0$. Together with the separated scalar a_0, this gives the inverse to $\Phi_{K,L}$. □

Padé reduction and multiplier-incidence geometry. For $F \in \mathcal{V}_{K,L}$, set

$$z_0 := A(0) + iB(0) \neq 0, \qquad C_F(u) + iS_F(u) := \frac{A(u) + iB(u)}{z_0}.$$

Then

$$C_F(0) = 1, \qquad S_F(0) = 0, \qquad \tan s_F(u) = \frac{S_F(u)}{C_F(u)} =: R_F(u).$$

Since A and B are polynomials, R_F is a rational function of bounded degree.

Proposition 9.3. *Fix (K, L) and define*

$$d_A := \max(2K - 2,\ 2L - 1), \qquad d_B := \max(2K - 1,\ 2L).$$

Then on $\mathcal{V}_{K,L}$, sufficiently high normalized Prüfer jets determine the rational phase quotient $R_F = S_F/C_F$ uniquely.

Proof. The formal series $s_F(u)$ determines the jet of $R_F(u) = \tan s_F(u)$ coefficientwise. Now write

$$R_F(u) = \frac{P(u)}{Q(u)}, \qquad Q(0) = 1, \qquad \deg P \le d_B, \quad \deg Q \le d_A.$$

The identity $Q(u)R_F(u) - P(u) = 0$ modulo $u^{d_A+d_B+2}$ gives the standard square Padé reconstruction system in the coefficients of P and Q, so sufficiently many coefficients of R_F determine the rational quotient uniquely. □

Theorem 9.4. *Fix a bounded bandwidth window (K, L) and an integer $s \geq 1$. Let*

$$R(u) = \frac{\widehat{S}(u)}{\widehat{C}(u)}$$

be a normalized rational phase quotient, where $(\widehat{C}, \widehat{S})$ *is the unique reduced coprime representative with* $\widehat{C}(0) = 1$ *and* $\widehat{S}(0) = 0$*. Define*

$$\mathcal{I}_s(K, L) := \left\{ (\widehat{C}, \widehat{S}, h) : \deg h = s,\ h(0) = 1,\ h(\widehat{C}, \widehat{S}) \text{ is realized in } \mathcal{V}_{K,L} \right\},$$

and let

$$\mathcal{Y}_s(K, L)$$

be its projection to the reduced-pair coordinates.

Then $\mathcal{I}_s(K, L)$ *is a real-algebraic subset of a finite-dimensional affine coefficient space, and* $\mathcal{Y}_s(K, L)$ *is a constructible, hence semialgebraic, subset of the reduced phase-pair parameter space.*

Proof. For fixed (K, L) and s, the reduced pair $(\widehat{C}, \widehat{S})$ has bounded degrees

$$\deg \widehat{C} \le d_A - s, \qquad \deg \widehat{S} \le d_B - s,$$

since multiplication by a degree-s polynomial must still fit inside the window bounds

$$\deg C \le d_A, \qquad \deg S \le d_B.$$

Choose coefficient coordinates for all such reduced pairs, together with coordinates for

$$h(u) = 1 + h_1 u + \cdots + h_s u^s.$$

The condition that $h(\widehat{C}, \widehat{S})$ be realized in $\mathcal{V}_{K,L}$ is a finite system of linear constraints on the coefficients of the product pair, hence a finite polynomial system in the coefficients of $\widehat{C}, \widehat{S}$, and h. Coprimeness of $(\widehat{C}, \widehat{S})$ is the nonvanishing of their resultant, hence a Zariski-open condition. Therefore $\mathcal{I}_s(K, L)$ is algebraic on the reduced locus, and its projection is constructible by elimination. □

Proposition 9.5. *Let* (C, S) *be a realized normalized phase pair on a bounded bandwidth window, and let*

$$d := \deg B$$

be the top degree of the underlying J_1*-channel. Write*

$$C(u) = \sum_{j=0}^{d} c_j u^j, \qquad S(u) = \sum_{j=1}^{d} s_j u^j.$$

Then the top two coefficient levels satisfy the universal quadratic relation

$$2(c_{d-1} s_d - s_{d-1} c_d) + c_d^2 + s_d^2 = 0.$$

*Moreover, this relation is covariant under degree-*1 *multiplication: if*

$$\widetilde{C} = (1 + tu)C, \qquad \widetilde{S} = (1 + tu)S,$$

then

$$Q_{d+1}(\widetilde{C}, \widetilde{S}) = t^2 Q_d(C, S),$$

where

$$Q_r(U, V) := 2(u_{r-1} v_r - v_{r-1} u_r) + u_r^2 + v_r^2$$

for

$$U(u) = \sum u_j u^j, \qquad V(u) = \sum v_j u^j.$$

Proof. Write the underlying realized pair as

$$F(t) = A(u)J_0(t) + B(u)J_1(t), \qquad x := A(0), \quad y := B(0), \quad D := x^2 + y^2.$$

If

$$A(u) = a\,u^{d-1} + \cdots, \qquad B(u) = b\,u^d + b'u^{d-1} + \cdots,$$

then

$$c_d = \frac{yb}{D}, \qquad s_d = \frac{xb}{D},$$

$$c_{d-1} = \frac{xa + yb'}{D}, \qquad s_{d-1} = \frac{xb' - ya}{D}.$$

Therefore

$$2(c_{d-1}s_d - s_{d-1}c_d) + c_d^2 + s_d^2 = \frac{2ab + b^2}{D}.$$

In the parity-adapted normalized Lommel basis, the highest coefficients always satisfy

$$2a + b = 0,$$

so the displayed expression vanishes.

For the covariance statement, if

$$\widetilde{C} = (1 + tu)C, \qquad \widetilde{S} = (1 + tu)S,$$

then

$$\widetilde{c}_{d+1} = tc_d, \qquad \widetilde{s}_{d+1} = ts_d,$$

$$\widetilde{c}_d = c_d + tc_{d-1}, \qquad \widetilde{s}_d = s_d + ts_{d-1}.$$

Substituting these formulas into Q_{d+1} gives

$$Q_{d+1}(\widetilde{C}, \widetilde{S}) = t^2 Q_d(C, S).$$

□

Proposition 9.6. *The first reduced-side degree-1 ambiguity loci can be computed exactly.*

(i) *In the smallest window* $(1, 0)$*, every reduced normalized phase pair has the form*

$$\widehat{C}(u) = 1 + pu, \qquad \widehat{S}(u) = qu, \qquad p, q \in \mathbb{R}.$$

Such a pair admits a nontrivial degree-1 lift into the window $(1, 1)$ *if and only if*

$$p^2 + (q + 1)^2 = 1.$$

(ii) *Every reduced normalized pair realized in the window* $(1, 1)$ *admits a nontrivial degree-1 lift into the window* $(2, 1)$*.*

Proof. For the window $(1,0)$, write

$$F = aJ_0 + b\widetilde{J}_2 + c\widetilde{J}_1, \qquad \widetilde{J}_2 = J_0 - 2uJ_1.$$

Then

$$A = a + b, \qquad B = c - 2bu.$$

With

$$x := A(0) = a + b, \qquad y := B(0) = c, \qquad D := x^2 + y^2,$$

the normalized pair is

$$C(u) = 1 - \frac{2by}{D}u, \qquad S(u) = -\frac{2bx}{D}u.$$

Hence every pair '(1+pu,qu)' is realized by choosing

$$x = q, \qquad y = p, \qquad b = -\frac{p^2+q^2}{2}, \qquad a = x - b, \qquad c = y.$$

Now multiply a reduced pair from $(1,0)$ by

$$h(u) = 1 + tu, \qquad t \neq 0.$$

The lifted coefficients are

$$C(u) = 1 + (p+t)u + pt\,u^2, \qquad S(u) = qu + qt\,u^2.$$

On the window $(1,1)$, Proposition 9.5 gives exactly the top-order relation

$$2(c_1s_2 - s_1c_2) + c_2^2 + s_2^2 = 0.$$

Substituting

$$c_1 = p + t, \qquad c_2 = pt, \qquad s_1 = q, \qquad s_2 = qt$$

gives

$$t^2\big(p^2 + (q+1)^2 - 1\big) = 0,$$

which proves part *(i)*.

For part *(ii)*, a normalized pair realized in $(1,1)$ has the form

$$\widehat{C}(u) = 1 + c_1u + c_2u^2, \qquad \widehat{S}(u) = s_1u + s_2u^2,$$

with

$$2(c_1s_2 - s_1c_2) + c_2^2 + s_2^2 = 0.$$

After multiplication by '1+tu', the lifted pair becomes

$$C(u) = 1 + (c_1+t)u + (c_2+tc_1)u^2 + tc_2\,u^3,$$

$$S(u) = s_1u + (s_2+ts_1)u^2 + ts_2\,u^3.$$

On the window $(2,1)$, Proposition 9.5 gives

$$2(c_2s_3 - s_2c_3) + c_3^2 + s_3^2 = 0.$$

Substituting the lifted coefficients yields

$$t^2\big(2(c_1s_2 - s_1c_2) + c_2^2 + s_2^2\big) = 0,$$

which vanishes identically because the original pair lies in the $(1,1)$ quadric. Hence every reduced normalized pair from $(1,1)$ admits a degree-1 lift into $(2,1)$. □

Proposition 9.7. *Fix a bounded bandwidth window $\mathcal{V}_{K,L}$ and let*

$$R(u) = \frac{\widehat{S}(u)}{\widehat{C}(u)}$$

be a normalized rational phase quotient, where $(\widehat{C}, \widehat{S})$ is the unique reduced coprime representative with $\widehat{C}(0) = 1$. Let

$$C_F + iS_F = \frac{A_F + iB_F}{z_{0,F}}$$

be the normalized phase pair of some $F \in \mathcal{V}_{K,L}$ realizing this quotient, and suppose

$$C_F = h\,\widehat{C}, \qquad S_F = h\,\widehat{S}$$

for a real polynomial h with $h(0) = 1$ and $\deg h = s \geq 0$. Then the top s coefficients of S_F vanish. More precisely, if

$$d_B := \max(2K - 1,\, 2L),$$

then

$$[u^{d_B}]S_F = [u^{d_B-1}]S_F = \cdots = [u^{d_B-s+1}]S_F = 0.$$

Proof. Write

$$h(u) = 1 + h_1 u + \cdots + h_s u^s, \qquad h_s \neq 0$$

if $s \geq 1$, and

$$\widehat{S}(u) = \sum_{r \geq 0} \sigma_r u^r.$$

Since $\deg S_F \leq d_B$, the identity

$$S_F = h\,\widehat{S}$$

implies that the coefficients of $u^{d_B+s}, u^{d_B+s-1}, \ldots, u^{d_B+1}$ in $h\,\widehat{S}$ vanish. The coefficient of u^{d_B+s} is $h_s \sigma_{d_B}$, so $\sigma_{d_B} = 0$. Proceeding downward, the coefficient of u^{d_B+s-r} forces $\sigma_{d_B-r} = 0$ for every $0 \leq r \leq s-1$. Since the top coefficients of S_F agree with those of $\widehat{S}$ shifted by the multiplier, this gives the claimed vanishing. □

The realized quadratic phase map. The realized-side normalized phase geometry is not linear. For

$$x = (A, B) \in \mathcal{V}_{K,L},$$

write

$$a := A(0), \qquad b := B(0), \qquad z_0 := a + ib.$$

Then the normalized phase pair is

$$C_x(u) + iS_x(u) := \frac{A(u) + iB(u)}{a + ib},$$

so

$$C_x(u) = \frac{aA(u) + bB(u)}{a^2 + b^2}, \qquad S_x(u) = \frac{aB(u) - bA(u)}{a^2 + b^2}.$$

Accordingly, the realized normalized phase quotient is

$$R_x(u) = \frac{Q_x(u)}{P_x(u)}, \qquad P_x(u) := aA(u) + bB(u), \qquad Q_x(u) := aB(u) - bA(u).$$

This is a self-normalized quadratic phase map in the realized coordinates.

Proposition 9.8. *For every bounded bandwidth window (K, L), the zero-phase fiber of the realized quadratic phase map is exactly*

$$\Psi_{K,L}^{-1}(0) = \mathbb{P}\langle J_0, J_1\rangle \cup \mathbb{P}\langle J_1, \widetilde{J}_2 - J_0\rangle.$$

Proof. If $R_x \equiv 0$, then $Q_x \equiv 0$. If $A(0) \neq 0$, this forces

$$B = \lambda A$$

for some real λ. The highest-degree triangular terms in the parity-adapted Lommel basis then eliminate every mode except the constant J_0 direction and the constant J_1 direction, so $x \in \mathbb{P}\langle J_0, J_1\rangle$.

If instead $A(0) = 0$, then admissibility implies $B(0) \neq 0$, and $Q_x \equiv 0$ forces

$$A \equiv 0.$$

The A-channel triangular reconstruction then leaves only the span of J_1 and $\widetilde{J}_2 - J_0$, which gives the second projective line. Both lines are contained in the fiber by direct substitution. □

Remark 9.9. The bounded-bandwidth inverse problem is therefore no longer a family-by-family elimination problem. After Theorems 9.1 and 9.2, the raw coefficient geometry is globally triangular, and after Proposition 9.3, the phase quotient is determined by a finite normalized Prüfer jet. On the reduced side, what remains is multiplier-incidence geometry, and Proposition 9.7 shows that every realized nonminimal lift is forced onto a successively thinner top-coefficient boundary stratum. On the realized side, the first global fiber is already explicit: Proposition 9.8 shows that the zero-phase fiber is the union of two canonical projective lines. Thus the bounded-bandwidth frontier has two complementary parts: a constructible incidence problem for reduced pairs, and a quadratic fiber geometry for realized pairs.

Appendix A. Verification of the low-order elimination formulas

The elimination arguments in Section 8 are all finite and explicit. More precisely, the anchored support-three theorems use only the low-order coefficients displayed in Propositions 8.7 and 8.11 and elementary algebraic operations on the resulting rational identities. In the mixed-parity and odd-odd sectors, the second-order contradiction is obtained by direct substitution

into the formulas for λ_1, λ_2. In the even-even sector, the fifth-order obstruction is reduced to the explicit quintic $\mathcal{P}_{M,N}(q)$, and the positivity arguments in Theorem 8.12 are then proved by the displayed derivative formulas, the adjacent-pair reduction, and the final Sturm count for the single exceptional polynomial $\mathcal{P}_{16,36}$. No computer-assisted search over parameter space is used in the proofs: the only symbolic input is the exact simplification of the low-order residuals into the displayed closed formulas.

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